What Is Stimulus?

Stimulus is a small tool.[33] Its own website describes it as “a modest JavaScript framework for the HTML you already have.” Where React tries to replace your entire view layer, Stimulus allows you to handle a common JavaScript pattern quickly and without much extra complication. That pattern is, “an event happens, and I’d like to update the DOM in response to it.”

Stimulus’s main concepts simplify the process of creating the relationship between an event and the code that gets invoked in response. If you weren’t using Stimulus, you’d likely try to handle these interactions with jQuery or just using native DOM methods without a framework at all. Stimulus, like Rails, was originally built by Basecamp, and has many convention-over-configuration design features that are reminiscent of the way Rails works.

Stimulus is particularly good for interactions that are relatively small and are expansions of existing browser capabilities. Stimulus is great at adding contained areas of interactivity to your application, like augmented forms, or simple navigation, and it works well as an enhancement to Turbo. We’re going to use Stimulus here to implement our Show/Hide button, and we’ll use it for filters and to support search within our page.

Stimulus lets you describe how your HTML markup relates to your JavaScript code. All of these relationships are inserted into your application by adding specific `data` attributes to DOM elements in the HTML markup that you send to the browser. When the page is loaded, Stimulus looks for these special attributes and uses them to set up event listeners behind the scenes. These listeners dispatch events to the controller method specified in your actions.

These are Stimulus’s core concepts:

Controllers
: Controllers are where your Stimulus JavaScript code goes. They contain methods that are called in response to events that are triggered in the